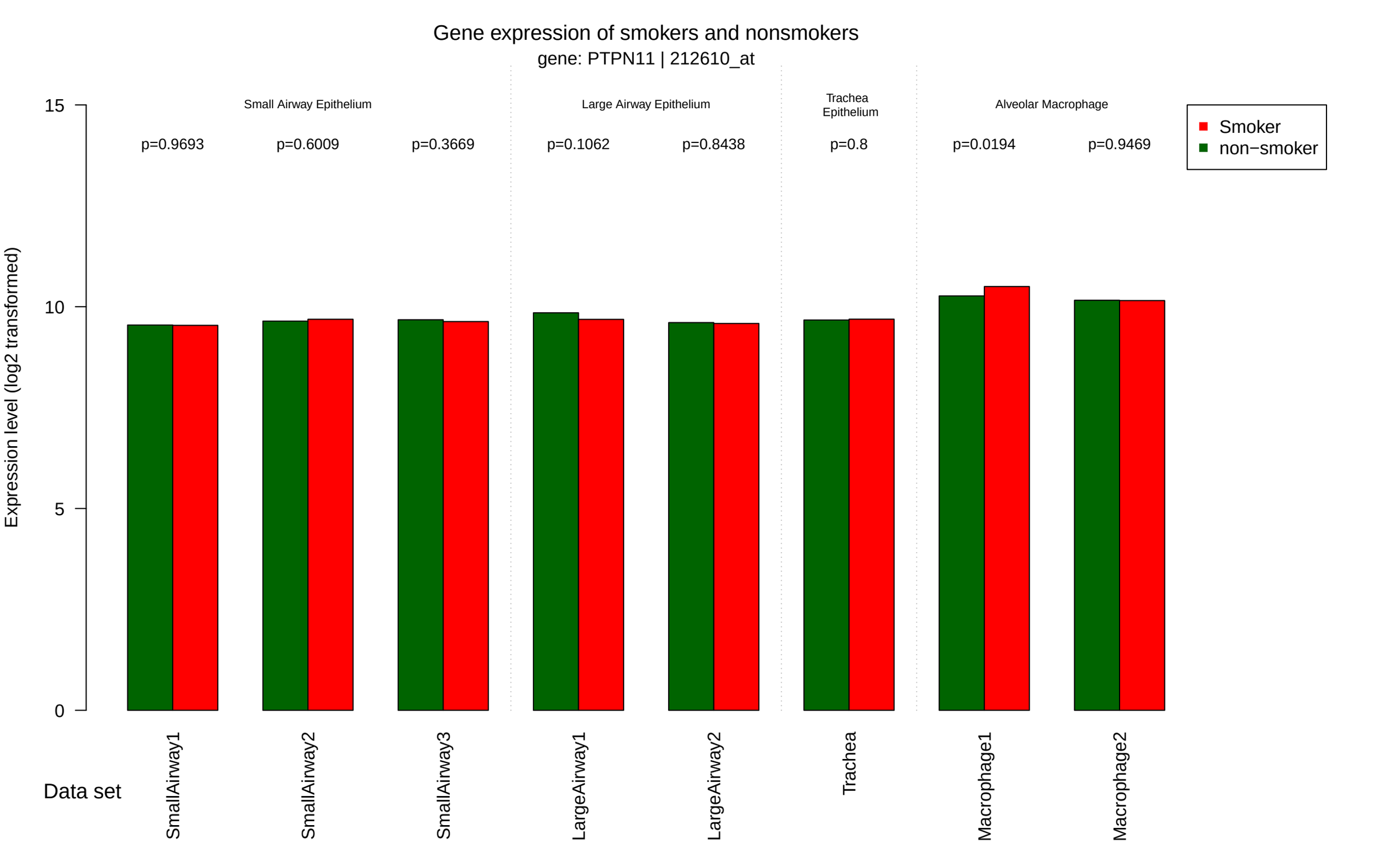
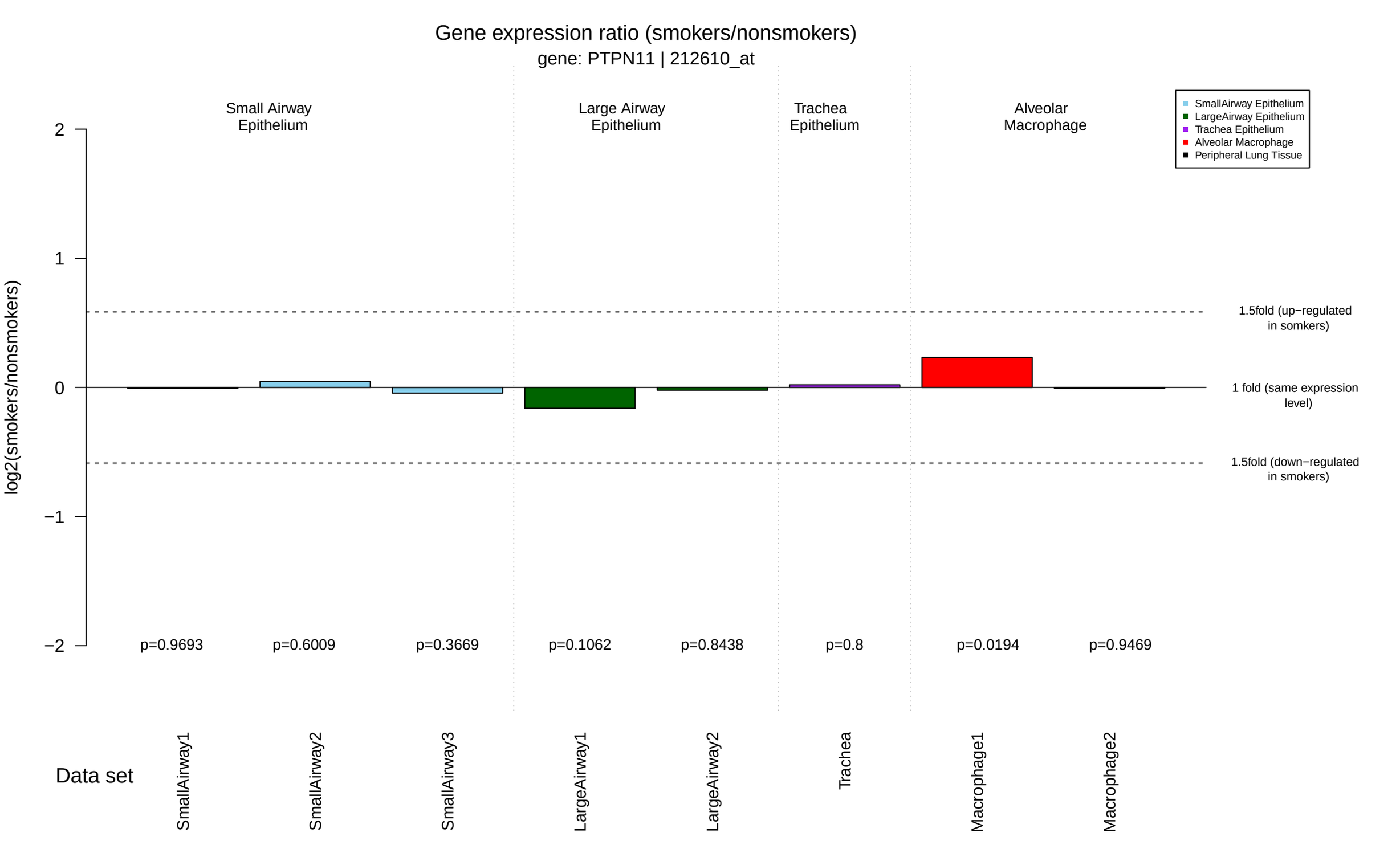
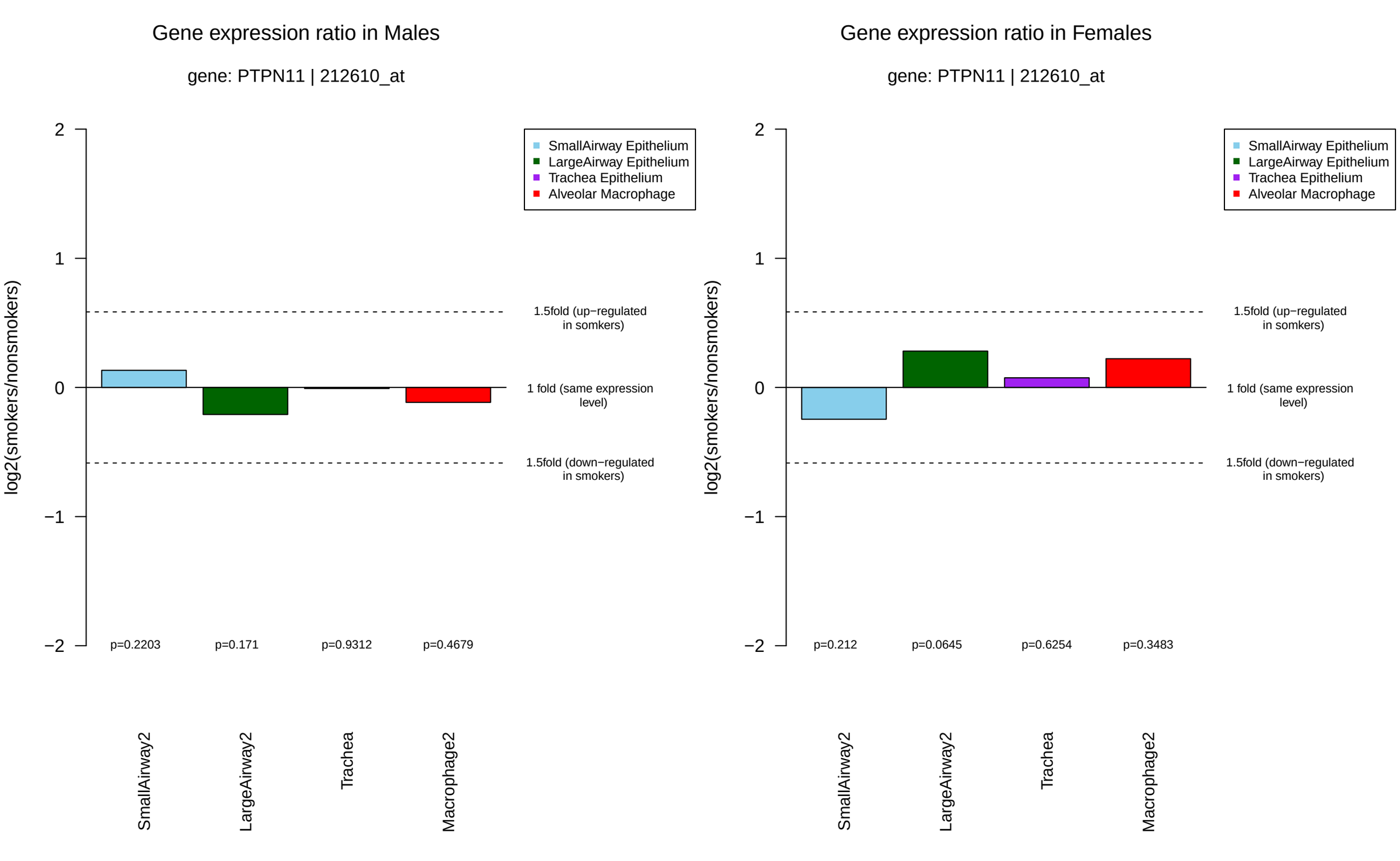
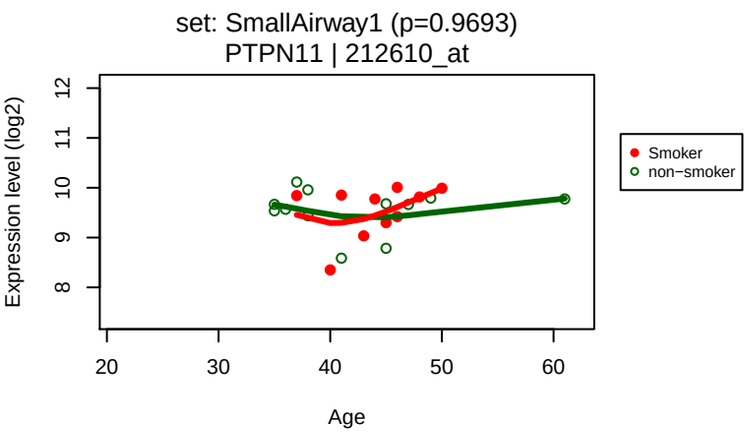
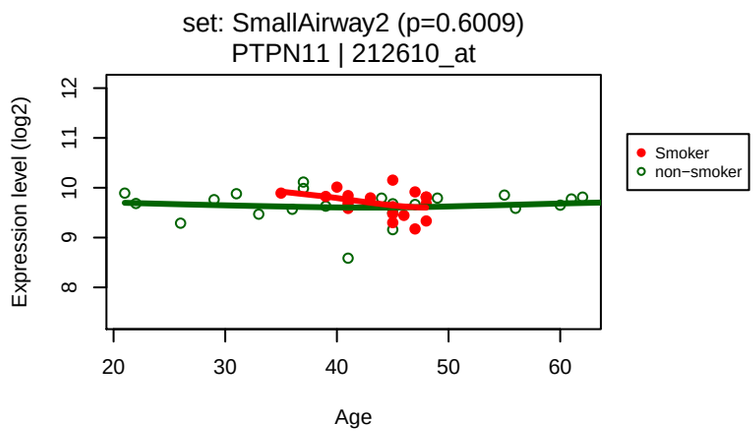
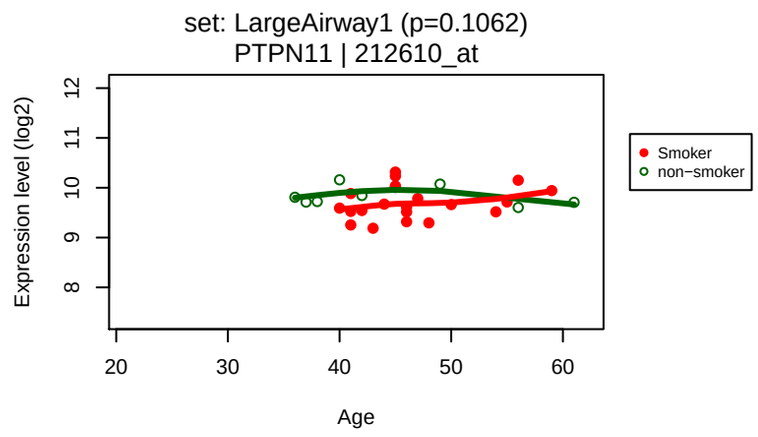

Gene expression of smokers and nonsmokers
gene: PTPN11 | 212610_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9693
p=0.6009
p=0.3669
p=0.1062
p=0.8438
p=0.8
p=0.0194
p=0.9469
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

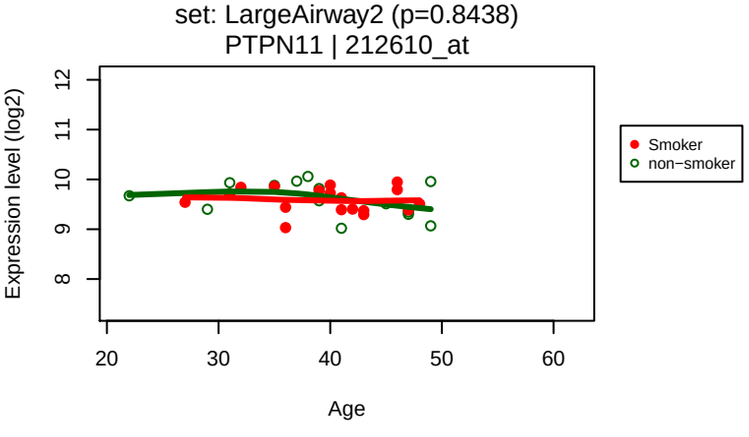

Gene expression ratio (smokers/nonsmokers)
gene: PTPN11 | 212610_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9693
p=0.6009
p=0.3669
p=0.1062
p=0.8438
p=0.8
p=0.0194
p=0.9469
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

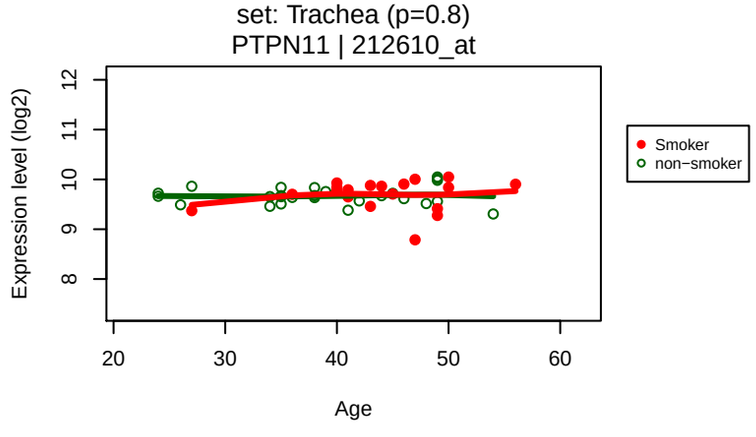

Gene expression ratio in Males
gene: PTPN11 | 212610_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2203
p=0.171
p=0.9312
p=0.4679
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN11 | 212610_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.212
p=0.0645
p=0.6254
p=0.3483
SmallAirway2
LargeAirway2
Trachea
Macrophage2

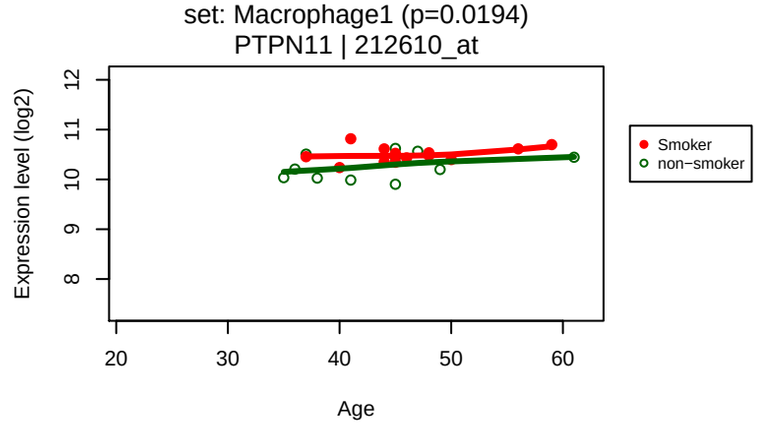

set: SmallAirway1 (p=0.9693)
PTPN11 | 212610_at
Expression level (log2)
Age
Smoker
non−smoker

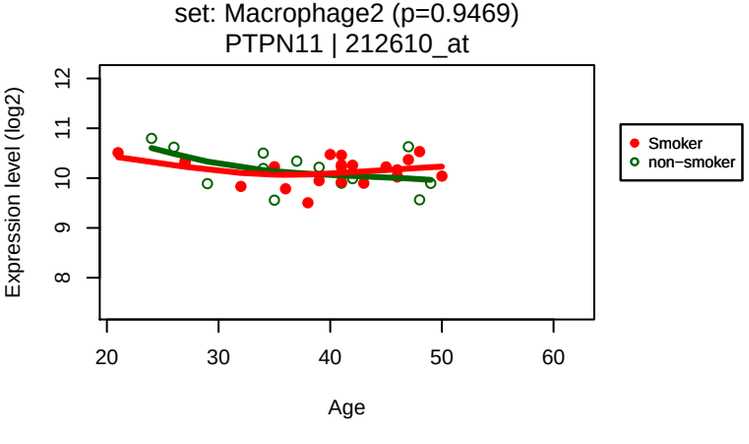

set: SmallAirway2 (p=0.6009)
PTPN11 | 212610_at
Expression level (log2)
Age
Smoker
non−smoker

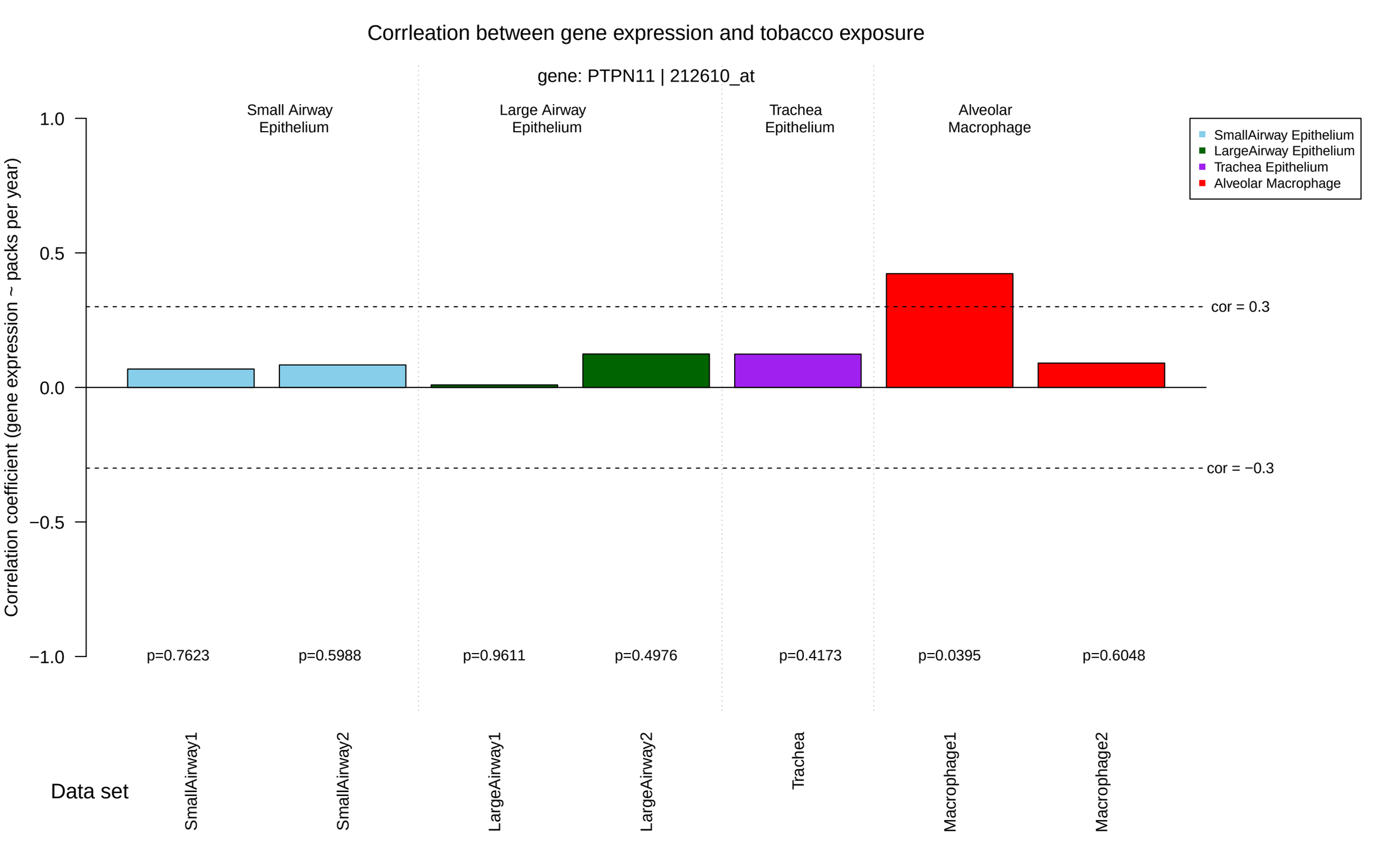

set: LargeAirway1 (p=0.1062)
PTPN11 | 212610_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8438)
PTPN11 | 212610_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8)
PTPN11 | 212610_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0194)
PTPN11 | 212610_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9469)
PTPN11 | 212610_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN11 | 212610_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7623
p=0.5988
p=0.9611
p=0.4976
p=0.4173
p=0.0395
p=0.6048
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2